Beyond Technique Hints

Can Alternative Memory Architectures Help on Math?

Date: 2026-03-21

Context: Follow-up to Headroom Search (2026-03-16)

Question: If technique-level concepts fail on math, does a fundamentally different memory type work?

Recap: Where We Left Off

The headroom search (2026-03-16) established:

Domain	Baseline	+ Technique Concepts	Delta
LiveCodeBench	80%	**85%**	+5
Math L5	98.5%	98.5%	0
Omni-MATH	**57.3%**	**52.3%**	-5.1
Omni d7-8	39%	22%	-17

Diagnosis: Concepts encode technique names ("use Vieta's formulas"). Math failures are reasoning depth, not knowledge gaps. The concept *type* is mismatched with the *failure mode*.

Open question: Would a fundamentally different memory architecture work?

Three Alternative Architectures Tested

Starting from the failure mode analysis, we evaluated four approaches:

ID	Approach	Mechanism	Status
I01+I03 → **I05**	Episodic memory (worked solutions)	Show full solution to similar problem	Tested
I06	Context warm-up (any solution)	Inject any math solution as priming	Tested
I04	Richer concept representation	More fields in concept schema	Vetoed (v3c precedent)
I02	Proof strategy hints	Higher-level hints	Parked (weak case)

Selection rationale: I05 had the strongest mechanistic case — episodic memory (full worked solutions) addresses reasoning depth differently than semantic memory (technique names). Supported by MemP finding that trajectories outperform scripts.

Phase 1: Retrieval Quality Audit

Question: Can embedding-based retrieval find structurally similar math problems?

Method: TF-IDF with bigrams over 4,428 Omni-MATH problems. 20 queries (d3-d7), two banks.

Retrieval Bank	Mean Top-1 Similarity
Math L5 → Omni query	0.330
Omni self (leave-one-out)	**0.485**

Retrieval quality depends on problem type

Problem Type	Typical Sim	Quality
Recurrences/sequences	0.7–0.8	Excellent
Functional equations	0.6–0.7	Good
Probability	0.5–0.9	Good
Infinite series	0.5–0.6	Good
Combinatorics	0.3–0.5	Mediocre
Geometry	0.2–0.4	Poor

Phase 2: Episodic Memory Smoketest (n=20)

Initial 2-condition result (looked promising)

Condition	Score	Delta
Baseline	16/20 (80%)	—
Episodic (TF-IDF relevant)	17/20 (85%)	+5pp

Zero regressions. Gained 1 problem (omath_0877, d=5.0).

At this point the signal looked identical to LCB (+5pp, zero regressions, 80% baseline).

The Random Control Changes Everything

The experiment-designer insisted on a 3rd condition: inject a **random** solved problem (not the most similar). This is the critical ablation — it tests whether the benefit comes from retrieval *relevance* or just extra *context*.

Condition	Score	Delta
Baseline	16/20 (80%)	—
Random example	**17/20 (85%)**	+5pp
Episodic (relevant)	17/20 (85%)	+5pp

Episodic = Random. Both gained +1 problem, but *different* problems:

- Random gained omath_2313, episodic gained omath_0877.
- The benefit was from extra context, not relevance.

Interpretation: Context Effect, Not Episodic Transfer

```
                    Retrieval relevance
                    does NOT matter
                          |
Baseline ──(+5pp)──> Random ──(0pp)──> Episodic
  80%                  85%                85%
                          |
                    Any worked solution
                    provides equal benefit
```

The +5pp was a **domain-general context warm-up** — seeing any mathematical reasoning primes the model. But this was n=20. Was the warm-up effect itself real?

Phase 3: Powered Warm-Up Experiment (n=108)

Scaled to 108 problems (12 per difficulty d1-d9). Added a **problem-only** control (random problem statement without solution — tests if the reasoning chain matters).

Condition	Score	Rate	vs Baseline
Baseline	**73/108**	**67.6%**	—
Math-warmup	71/108	65.7%	-1.9pp
Problem-only	72/108	66.7%	-0.9pp

The warm-up effect was noise. All three conditions within 2pp. Both treatment conditions scored *below* baseline.

Per-Difficulty Breakdown (n=108)

Diff	Baseline	Math-warmup	Problem-only
d1	83%	75%	75%
d2	92%	92%	100%
d3	83%	83%	75%
d4	67%	67%	67%
d5	58%	42%	58%
d6	67%	58%	58%
d7	50%	58%	42%
d8	50%	58%	58%
d9	58%	58%	67%

No consistent advantage at any difficulty level. Random variation dominates.

The Smoketest Trap

The n=20 smoketest showed +5pp (80% → 85%). This looked identical to the validated LCB result. But:

Stage	n	Apparent Effect	Real Effect
Smoketest (2 conditions)	20	+5pp	Unknown
Smoketest + random control	20	+5pp (both)	Context effect?
Powered experiment	108	0pp	**Null**

Lesson: A single +1/20 flip is a coin toss (p=0.50 by McNemar's test). The experiment guardrail "do not conclude from fewer than 50 samples" exists for exactly this reason.

The random control was essential. Without it, we would have scaled the episodic pipeline to n=225 before discovering the null. The control saved ~$10 and several hours by collapsing the hypothesis at n=20.

Why Math Is Structurally Immune

Every tested memory architecture fails on math. The pattern:

Architecture	What It Provides	What Math Needs	Result
Technique concepts	"Use Vieta's formulas"	Multi-step reasoning	-5pp
Relevant worked solution	Full solution to similar problem	Creative insight	0pp
Random worked solution	Any math reasoning chain	Novel approach path	0pp
Problem statement only	Priming with math problem	Deep combinatorial search	0pp

The model already has technique knowledge and can follow worked solutions. Its failures on olympiad math are:

1. **Execution errors** (d1-d4): arithmetic mistakes, missed edge cases — not addressable by hints
2. **Reasoning depth** (d5-d7): needs 4-5 step chains without error — no external hint fixes this
3. **Creative insight** (d8-d9): needs non-obvious constructions — hints push wrong paths

Cross-Domain Comparison

These math experiments complete the picture across all tested domains:

Domain	Benchmark	Baseline	Memory Type	Effect	N
Code	LCB	80%	Technique concepts	**+5pp**	100
Math (competition)	Math L5	98.5%	Technique concepts	0	200
Math (olympiad)	Omni-MATH	57%	Technique concepts	-5.1	225
Math (olympiad)	Omni-MATH	68%	Episodic (relevant)	0	108
Math (olympiad)	Omni-MATH	68%	Context warm-up	0	108
Science	GPQA Diamond	83%	Explanation injection	+0.5	200
Tool use	BFCL-V4 (exec)	91%	—	ceiling	150

The Failure Mode Taxonomy

The key insight: **memory augmentation is not universally helpful.** It helps if and only if the failure mode is a knowledge gap.

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Failure Mode          Memory Helps?     Example

Knowledge gap         YES               LCB: model doesn't know
(model lacks info)                       itertools.accumulate pattern

Reasoning depth       NO                Omni-MATH: model knows
(model can't chain)                     techniques but can't chain
                                        4 steps without error

Ceiling               NO                Math L5 (98.5%),
(model already wins)                    BFCL exec (91%)

Creative insight      HARMFUL           Omni d7-8: hints push
(needs novel idea)                      plausible but wrong paths
```

What Worked in the Process

1. The random control saved the project

At n=20, episodic memory looked like a breakthrough (+5pp, zero regressions). The random-example control — suggested by the experiment-designer agent — immediately showed the effect was domain-general context, not retrieval-specific. This prevented weeks of wasted pipeline work.

2. Retrieval quality audit as a gate

Before building any pipeline, we tested whether TF-IDF could find structurally similar math problems. This cheap check (zero API cost) validated the retrieval mechanism before committing to implementation.

3. Progressive sample sizes

Each stage increased commitment only after passing the previous gate:

Stage	Cost	What It Tested	Gate Passed?
Retrieval audit	$0	Can we find similar problems?	Yes (50% of types)
2-condition smoketest	~$1	Does it work at all?	Yes (misleadingly)
3-condition smoketest	~$0.50	Is it relevance or context?	No (context only)
Powered experiment	~$3	Is the context effect real?	No (null)

What Failed in the Process

1. Trusting n=20

The +5pp smoketest result triggered excitement before it was validated. The research principles state "positive results are a trigger to validate harder" — but the natural reaction to +5pp with zero regressions is to celebrate, not to add more controls.

2. Initial design missing the random control

The first smoketest had only baseline and episodic conditions. The experiment-designer agent caught this gap. Without mandatory experiment-designer dispatch, the random control would not have been run at the smoketest stage.

3. Option shuffling variance

On GPQA, baseline accuracy varied 4pp between seeds (81% → 85%) purely from shuffling answer option positions. This is a known MCQ evaluation issue — answer position affects model behavior. Single-seed MCQ results are unreliable.

Recommendations

For math: direction is closed

Every tested memory architecture fails. Further math gains require a fundamentally different approach:

- **Process-level improvements** (retry strategies, self-correction)
- **Different problem decomposition** (break into subproblems, verify steps)
- **Training-time interventions** (not inference-time memory augmentation)

For the broader concept memory project

The validated finding remains: **concept memory helps when failure mode = knowledge gap (LCB +5pp)**. This is narrow but real. The path forward:

1. **Scale the LCB result** to 300+ problems for statistical confidence
2. **Test more knowledge-gap domains** (GPQA showed null but n was small; BFCL needs harder splits)
3. **Frame honestly**: concept memory is a domain-specific tool for knowledge-gap failure modes, not a universal augmentation

For experimental methodology

- Always include a random-example control when testing retrieval/selection

Method Evolution DAG

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A01 Pipeline architecture (Assumed)
├── A05 LCB concept benefit +5pp (Assumed)
│   ├── I07 GPQA Diamond (exploring → null at 2 seeds)
│   └── I08 BFCL-V4 (exploring → ceiling on available splits)
├── A06 Technique concepts neutral/harmful on math (Assumed)
│   └── A07 Failure mode taxonomy (Assumed)
│       ├── I05 Episodic memory (NEGATIVE — relevance = random)
│       │   ├── A08 TF-IDF retrieval viable for formulaic types (Assumed)
│       │   └── I06 Context warm-up (NEGATIVE — null at n=108)
│       ├── I02 Proof strategy hints (Parked — weak case)
│       └── I04 Richer representation (Vetoed — v3c precedent)
└── A02 Two-stage concept extraction (Assumed)
```

16 nodes: 8 Assumed, 2 exploring, 2 negative, 1 parked, 3 vetoed.

Summary

Question: Can alternative memory architectures help on math?

Answer: No. We tested three fundamentally different approaches beyond technique concepts:

1. **Episodic memory** (relevant worked solutions): 0pp — relevance doesn't matter
2. **Context warm-up** (any worked solution): 0pp — smoketest noise
3. **Explanation injection on GPQA**: +0.5pp — null across seeds

Math failures are reasoning depth, not knowledge gaps. No form of external knowledge injection at inference time addresses this. The concept memory framework is validated for knowledge-gap domains (LCB +5pp) and only knowledge-gap domains.

Cost of this investigation: ~$8 in API calls, 3 sessions of work.

Key methodological contribution: The random-example control pattern. Should be mandatory for all retrieval/selection experiments.